



Research paper

Robust oxygen excess ratio control of PEMFC systems using adaptive dynamic programming[☆]Jing Zhu^{*}, Peng Zhang, Xiang Li, Bin Jiang

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ABSTRACT

To improve the power generation efficiency and cells lifetime of proton exchange membrane fuel cells, the oxygen excess ratio (OER) control problem is investigated in this paper. With the use of the double-loop cascade control structure, our work reinforces the iteration adaptive dynamic programming (ADP) algorithm and develops a robust OER control strategy drawing upon rich tools and techniques in sliding mode control theory. Our contribution is threefold. First, the proposed OER control policy is insensitive to various dynamics and disturbances by using refined super-twisting sliding mode controller in the external control loop. Second, the ADP-based controller is employed in the internal control loop to approximate the real optimal tracking control policy. The algorithm is computationally efficient and easy to implement in hardware platforms. Third, simulation results show that the proposed OER control strategy has a quick, smooth response, which can be further refined by means of numerous advanced neural network algorithms.

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1. Introduction

The global energy shortage has been a critical issue in modern society. With the increasing contradiction between energy demands and energy reserves, more and more attention has been paid to the development of renewable energy like solar energy, wind power, bioenergy, etc. Hydrogen fuel cell, which converts chemical energy to electrical energy directly, therefore, draws a steadily growing interest among the world by virtue of high energy density, low construction cost and zero air pollution. An extraordinary volume of developments has been made on hydrogen energy strategies, regulations schemes and operation optimizations (see Midilli et al., 2004; Bianchi et al., 2013; Hasikos et al., 2009, and the references therein). Among the large work, the majority are of the advances on proton exchange membrane fuel cells (PEMFCs). As the most widely used hydrogen energy apparatus, PEMFC systems have the significant attributes of high efficiency, high reliability, low working noise, and flexible modular structure, hence are largely used in military spaces, unmanned aerial vehicles, cars, and mobile devices, to name a

few (Westbrook, 2001; Herwerth et al., 2007; Zhu et al., 2020b; Bruck et al., 2004; Bruijn, 2005).

The structure of PEMFC system is as depicted in Fig. 1, consisting of PEMFC stacks and four main subsystems, that is hydrogen supply subsystem, air delivery subsystem, humidity management subsystem, and temperature regulation subsystem. Electrical energy is generated in FC stacks through electrochemical reaction with adequate supplies of hydrogen and air (oxygen) from Cathode and Anode. To guarantee the high efficiency of power generation, it is necessary to regulate the air flow in air delivery subsystem such that the oxygen excess ratio (OER) is within the certain range. Otherwise, the oxygen starvation phenomenon occurs when the OER is lower than one, giving rise to cell voltage reduction, stack flooding, membrane calorification, and even membrane damage (Pukrushpan et al., 2004; Yang et al., 2012; Abdullah and Idres, 2014). Excessive power is consumed by air compressor and the power generation efficiency is degraded when the OER is too large. The optimal performance is achieved when OER is around 2.05 after numerous experiments. Therefore, OER control is of considerable importance in PEMFC systems from both economic and environmental perspectives.

Given the nonlinear properties of PEMFC systems, the majority of achievements on OER control are proposed based upon linearization techniques. The eighth-order linearized system is adopted in Pukrushpan et al. (2004) to put forward the hybrid control policy by static feedforward (SFF) and observer based feedback controllers. Linear quadratic gaussian (LQG) controller is adopted for OER regulation under load fluctuation in Niknezhadi

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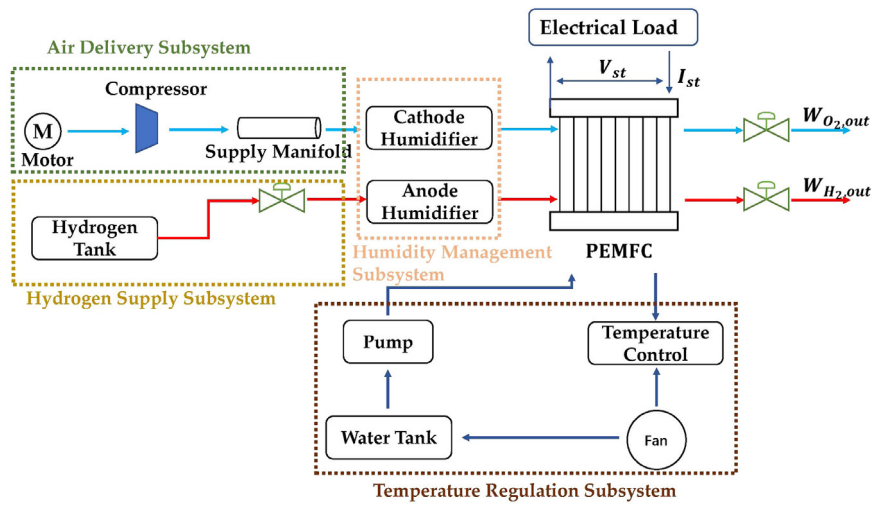


Fig. 1. The overall structure of PEMFC systems.

et al. (2011). The passivity PI controller for predefined local linearized model is used in Talj et al. (2011). On the other hand, intelligent control strategy in terms of fuzzy control has been largely applied in the OER control problem. Ou et al. (2015) and Baroud et al. (2016) propose the feedforward and hybrid fuzzy-PID controller for air flow regulation in PEMFC, respectively. Beirami et al. (2015) proposes the optimal PID-fuzzy control scheme by using evolution algorithm. These fuzzy controllers have been successfully used in PEMFC air feed system with the advantage of strong robustness. However, the quantification and scale factors in fuzzy controller generally depend on expert experience, giving rise to insufficient control accuracy and stubborn steady-state error (Zhu et al., 2020a; Baroud et al., 2017; Tang et al., 2021). A satisfactory answer is yet to be available until now.

To cope with the uncertainties in PEMFC, adaptive nonlinear controller is applied in Bao et al. (2006) along with LQG controller to improve the robustness, and the linear parameter varying (LPV) gain-scheduled control scheme is proposed concerning the LPV anti-windup compensation under control constraints. In work (Kazmi and Bhatti, 2012; Kim, 2012), the sliding mode observer is used to estimate nonlinear model parameter for robust OER control. To addressing the chattering at high-frequency switching Liu et al. (2016), Rakhtala et al. (2015, 2014), Piloni et al. (2015), Deng et al. (2017) introduce the high-order sliding mode (HOSM) algorithm to realize OER estimation and control.

Our objective in this paper is to drive the OER to the optimal value quickly and robustly in terms of adaptive dynamic programming algorithm. The idea of adaptive dynamic programming (ADP) is first advocated to solve the Hamilton–Jacobi–Bellman (HJB) equation in Werbos (1977). Afterwards, ADP has distinguished itself for its effectiveness in dealing with the notorious curse of dimensionality phenomenon (Grüne and Semmler, 2004). Among several branches of ADP algorithms, iteration ADP algorithm exhibits the best universality in both continuous-time and discrete-time optimal control problem. Our work reinforces the iteration ADP algorithm to develop a robust OER control strategy. In this vein, how to set up the nonlinear mathematical expression with fictitious modeling uncertainties remains crucial. To this effect, a key step is to formulate the nonlinear model of PEM fuel cell systems against fictitious modeling uncertainties. We adopt the double-loop cascade structure (Werbos, 1977), in which the external control loop generates the dynamic reference trajectory on air compressor’s angular velocity and the internal regulation loop calculates the optimal voltage signal. More specifically, the actual OER is driven to the optimal value in the external control

loop with the use of sliding mode control, while the optimal voltage signals are calculated iteratively in the internal loop based on ADP algorithm (Zhang et al., 2011; Song et al., 2014). Our contribution is threefold. Firstly, we provide a robust OER control policy which is insensitive to various dynamics and disturbances in PEMFC by using improved super-twisting sliding mode controller. Next, with the use of neural networks (NNs) and ADP algorithm, the proposed control strategy is easy to implement in hardware platform and the computational complexity is severely reduced. Finally, our proposed OER control strategy demonstrates smooth and quick response and can be improved further by means of numerous advanced ADP techniques. Simulation results show that the proposed control policy is robust and has a chatter-free voltage input when the load current varies by virtue of super-twisting controller.

The rest of the paper is organized as follows. The nonlinear mathematical expression of the PEMFC system along with the double-loop control structure are illustrated in Section 2. In Section 3, the robust OER control strategy is proposed, which consists of the super-twisting based controller in the external loop and the ADP based controller in the internal loop. The efficiency of the proposed control strategy is proved theoretically. Simulation results are presented in Section 4 to show the effectiveness of the proposed OER control scheme compared with other OER control scheme. Finally, conclusions are given in the end.

2. Mathematical model

In this section, we shall introduce the nonlinear mathematical model of PEMFC systems. Let us begin with the description on notations. For any nonsingular matrix M , we denote its transpose by M^T and inverse by M^{-1} . The least and largest eigenvalues of M are denoted by $\rho_{\min}(M)$ and $\rho_{\max}(M)$. For any stable function $f(x)$, we denote its transpose by $f^T(x)$ and inverse by $f^{-1}(x)$. The Euclidean norm of $f(x)$ is denoted by $\|f(x)\|$, which is defined as

$$\|f(x)\| = (f^T(x)f(x))^{1/2}. \quad (1)$$

We assume that the pressures of the cathode, supply manifold, and angular speed are measurable. In order to avoid frequency measurement error and signal fluctuation, we adopt an advanced fourth-order mathematical model in Laghrouche et al. (2015) and Talj et al. (2009a)

$$\dot{x} = \mathcal{G}(x) + \Phi \cdot u_{cm} + \Gamma \cdot I_{st}, \quad (2)$$

Table 1
Key variables in PEMFCs.

Symbol	Unit	Explanation
p_{ca}	Pa	Cathode pressure
p_{N_2}	Pa	Nitrogen pressure
ω_{cp}	rad/s	Angular velocity of the compressor
p_{sm}	Pa	Supply manifold pressure
I_{st}	A	Load current
u_{cm}	V	Voltage
p_{net}	kPa	Net power
λ_{O_2}	–	Oxygen excess ratio
$W_{ca,out}$	Pa	Cathode flow rate
W_{cp}	Pa	Compressor flow rate
$W_{O_2,ca,in}$	Pa	Cathode oxygen flow rate
$W_{O_2,react}$	Pa	Reacting oxygen in fuel cell stack

with system states being $x = [x_1, x_2, x_3, x_4]^T = [p_{ca}, p_{N_2}, \omega_{cp}, p_{sm}]^T$. The physical meaning of p_{ca} , p_{N_2} , ω_{cp} , p_{sm} are addressed in [Table 1](#). The output of the system is $y = [x_1, x_3, x_4]^T$. The nonlinear function $\mathcal{G}$ is in the following form

$$\mathcal{G}(x) = \begin{pmatrix} -\frac{b_3(x_1 - b_2)}{nx_1} W_{ca,out} - (b_1 + b_5)(x_1 - x_4) \\ -\frac{b_3 x_2}{nx_1} W_{ca,out} - b_5(x_1 - x_4) \\ -\frac{b_7}{x_3} \beta W_{cp} - b_6 x_3 \\ b_{11}(1 + b_{12}\beta) \cdot (W_{cp} + b_{13}(x_1 - x_4)) \end{pmatrix}, \quad (3)$$

where $\beta = (x_4^{b_9} - b_8^{b_9})/b_8^{b_9}$, b_i for $i = 1, \dots, 16$, n are constant values as defined in [Tables 2](#) and [3](#). $W_{ca,out}$ is the cathode flow rate with

$$W_{ca,out} = k_{ca,out} \sqrt{p_{ca} - p_{atm}}, \quad (4)$$

where $k_{ca,out}$, p_{atm} are constants given in [Table 3](#). The compressor flow rate W_{cp} is computed by [Talji et al. \(2009b\)](#) as $W_{cp} = b_{14}\omega_{cp}$. Φ and Γ are vectors with

$$\Phi = \begin{pmatrix} 0 \\ 0 \\ b_{10} \\ 0 \end{pmatrix}, \quad (5)$$

and

$$\Gamma = \begin{pmatrix} b_4 \\ 0 \\ 0 \\ 0 \end{pmatrix}. \quad (6)$$

Let λ_{O_2} , $\lambda_{O_2}^*$ denote the actual and optimal OER value, respectively, where

$$\lambda_{O_2} = \frac{W_{O_2,ca,in}}{W_{O_2,react}} = \frac{b_{15}(x_4 - x_1)}{b_{16}I_{st}}, \quad (7)$$

and $\lambda_{O_2}^*$ is 2.05 ([Rakhtala et al., 2015](#)).

To improve the regulation response and robustness of OER control, we use the double-loop cascade control structure depicted in [Fig. 2](#).

- 1. The External Control Loop:** The objective of the external control loop is to turn λ_{O_2} into $\lambda_{O_2}^*$. To this effect, we employ the HOSM controller to get the optimal angular velocity ω_{cp}^* . The use of super-twisting and SFF controller renders the control scheme robust against uncertainties and disturbances. Moreover, chattering is transferred from low frequency to high frequency by choosing appropriate parameters in the super-twisting controller.

Table 2
Values of parameter b_i for $i = 1, \dots, 16$.

$b_1 = \frac{RT_{fc} k_{ca,in}}{M_{O_2} V_{ca}} \left(\frac{x_{O_2,atm}}{1 + \omega_{atm}} \right)$	$b_2 = p_{sat}$
$b_3 = \frac{RT_{fc}}{V_{ca}}$	$b_4 = \frac{nRT_{fc}}{4FV_{ca}}$
$b_5 = \frac{RT_{fc} k_{ca,in}}{M_{N_2} V_{ca}} \left(\frac{1 - x_{O_2,atm}}{1 + \omega_{atm}} \right)$	$b_6 = \frac{f}{J_{cp}}$
$b_7 = \frac{C_p T_{atm}}{J_{cp} \eta_{cp}}$	$b_8 = p_{atm}$
$b_9 = \frac{\gamma - 1}{\gamma}$	$b_{10} = \frac{\eta_{cm} k_t}{J_{cp}}$
$b_{11} = \frac{\gamma RT_{atm}}{M_a V_{sm}}$	$b_{12} = \frac{1}{\eta_{cp}}$
$b_{13} = k_{sm,out}$	$b_{14} = \frac{0.85}{2\pi} V_{cp/tr} \rho_a$
$b_{15} = k_{ca,in} \frac{x_{O_2,atm}}{1 + \omega_{atm}}$	$b_{16} = \frac{nM_{O_2}}{4F}$

- 2. The Internal Control Loop:** The objective of the internal control loop is to turn ω_{cp} into ω_{cp}^* in terms of optimal tracking controller. The ADP based controller is applied to approximate the actual optimal controller and consequently reduces the computation complexity and alleviates the chattering phenomenon efficiently. The weights of NN are online tuned and the estimate errors converge to zero.

3. Main results

For most practical applications of PEMFCs, load current fluctuation is a common scenario which may cause the variation of OER. In this section, we develop the robust OER control strategy such that the actual OER is driven to the optimal value rapidly and smoothly.

3.1. Super-twisting control algorithm

Let $r(t)$ denote the difference between λ_{O_2} and $\lambda_{O_2}^*$ with

$$r(t) = \lambda_{O_2} - \lambda_{O_2}^*. \quad (8)$$

Since that $\lambda_{O_2}^*$ is constant, the variation rate of $r(t)$ is calculated as

$$\dot{r}(t) = \dot{\lambda}_{O_2} = \frac{b_{15}}{b_{16}} \left(\frac{x_4 - x_1}{I_{st}} \right)', \quad (9)$$

which is further equivalent to

$$\dot{r}(t) = \xi_1(x, t) + \xi_2(x, t)x_3, \quad (10)$$

with

$$\xi_1(x, t) = \frac{b_{15}}{b_{16}I_{st}^2} \cdot M, \quad (11)$$

$$\xi_2(x, t) = \frac{b_{11}b_{14}b_{15}}{b_{16}I_{st}} (1 + b_{12}\beta), \quad (12)$$

and

$$M = \left(b_4 I_{st} - (b_1 + b_5)(x_4 - x_1) - \frac{b_3(b_2 - x_1)}{nx_1} \cdot W_{ca,out} \right) I_{st} - (I_{st} + b_{11}b_{13}(1 + b_{12}\beta)(x_4 - x_1)). \quad (13)$$

It is useful to make the following assumptions to simplify the sliding mode controller synthesis.

Assumption 1. Load current I_{st} is continuously differentiable and satisfies

$$|I_{st}| \leq I_1, \quad |\dot{I}_{st}| \leq I_2. \quad (14)$$

where I_1 and I_2 are some positive constants.

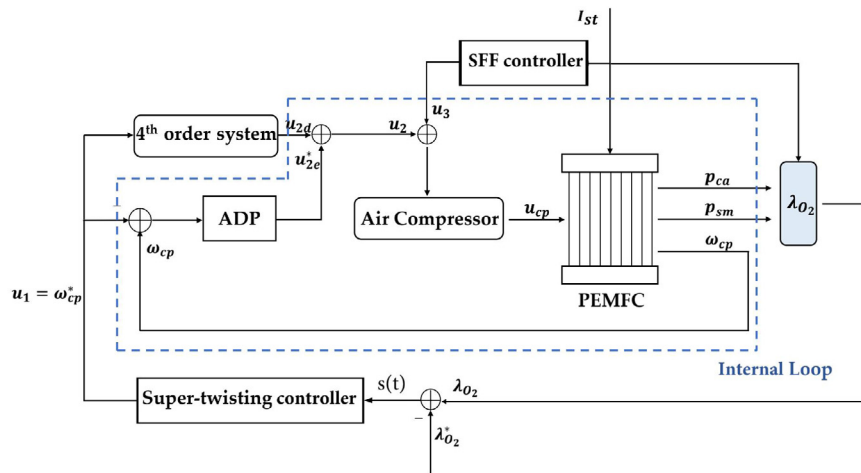


Fig. 2. Double-loop cascade structure for OER control.

Assumption 1 guarantees that the load current I_{st} does not vary intensely. In practical PEMFC applications, the current rate limiter is generally used as an auxiliary equipment to avoid big spikes (Chen et al., 2018).

Assumption 2. $\xi_1(x, t)$ and $\xi_2(x, t)$ in (11) and (12) are bounded by

$$|\xi_1(x, t)| \leq \bar{\varepsilon}_1, \quad (15)$$

$$0 < \underline{\varepsilon}_2 < \xi_2(x, t) < \bar{\varepsilon}_2, \quad (16)$$

where $\bar{\varepsilon}_1$, $\underline{\varepsilon}_2$, and $\bar{\varepsilon}_2$ are positive constants.

Assumption 2 is essential to make sure that the state $r(t)$ in (10) is stabilizable in finite-time. Note that herein we assume that ξ_2 is positive and bounded. Consequently, we have the following bound

$$\dot{r}(t) \in [-\bar{\varepsilon}_1, \bar{\varepsilon}_1] + [\underline{\varepsilon}_2, \bar{\varepsilon}_2] \omega_{cp}. \quad (17)$$

In the case that $\xi_2(x, t)$ is negative, the similar bound is derived by changing the sign of x_3 . In fact, the exact knowledge of $\bar{\varepsilon}_1$, $\underline{\varepsilon}_2$, $\bar{\varepsilon}_2$ is not necessary in practice since that they only have impact on the magnitude of proposed controller (see pp. 216–217, Shtessel et al., 2014).

In terms of super-twisting control methods, we are led to the following theorem.

Theorem 1. Consider the cascade-control structure in Fig. 2. The input of sliding mode controller $r(t)$ is convergent to zero within finite time T_f if the following super-twisting controller is adopted

$$u(t) = -p_1 \text{sign}(r(t)) \sqrt{|r(t)|} - p_3 r(t) - p_2 \int_0^t \text{sign}(r(s)) ds - p_4 \int_0^t r(s) ds. \quad (18)$$

The tuning parameter p_i , for $i = 1, \dots, 4$, is

$$\begin{cases} p_1 = \tilde{p}_1 \sqrt{H}, \\ p_2 = \tilde{p}_2 H, \\ p_3 = \tilde{p}_3 H, \\ p_4 = \tilde{p}_4 H^2, \end{cases} \quad (19)$$

respectively, where $\tilde{p}_1$, $\tilde{p}_2$, $\tilde{p}_3$, $\tilde{p}_4$ and H are positive constants satisfying

$$\frac{4\tilde{p}_2\tilde{p}_3^{-2}\tilde{p}_4}{9\tilde{p}_1^2 + 8\tilde{p}_2} > 1. \quad (20)$$

Proof. Let $r_1 = r(t)$, and $r_2 = -p_2 \int_0^t \text{sign}(r(s)) ds - p_4 \int_0^t r(s) ds$.

The super-twisting control policy (18) is equivalent to the following expression

$$\begin{aligned} \dot{r}_1 &= -p_1 \sqrt{|r_1|} \text{sign}(r_1) - p_3 r_1 + r_2, \\ \dot{r}_2 &= -p_2 \text{sign}(r_1) - p_4 r_1. \end{aligned} \quad (21)$$

Construct a continuous and differentiable Lyapunov function

$$V_1(x) = x^T Q x, \quad (22)$$

where Q is symmetric positive definite matrix with appropriate dimension, and

$$x = \begin{pmatrix} \sqrt{H|r_1|} \text{sign}(r_1) \\ H r_1 \\ r_2 \end{pmatrix}. \quad (23)$$

Let Q be equal to

$$Q = 0.5 \begin{pmatrix} 4\tilde{p}_2 + \tilde{p}_1^2 & \tilde{p}_1\tilde{p}_3 & -\tilde{p}_1 \\ \tilde{p}_1\tilde{p}_3 & \tilde{p}_3^2 + 2\tilde{p}_4 & -\tilde{p}_3 \\ -\tilde{p}_1 & -\tilde{p}_3 & 2 \end{pmatrix}. \quad (24)$$

The derivative of $V(x)$ in (22) is consequently described as

$$\dot{V}_1(x) = -|r_1|^{-\frac{1}{2}} x^T Q_1 x - x^T Q_2 x, \quad (25)$$

where

$$Q_1 = \frac{\tilde{p}_1}{2} \begin{pmatrix} \tilde{p}_1^2 + 2\tilde{p}_2 & 0 & -\tilde{p}_1 \\ 0 & 5\tilde{p}_3^2 + 2\tilde{p}_4 & -3\tilde{p}_3 \\ -\tilde{p}_1 & -3\tilde{p}_3 & 1 \end{pmatrix}, \quad (26)$$

and

$$Q_2 = \tilde{p}_3 \begin{pmatrix} 2\tilde{p}_1^2 + \tilde{p}_2 & 0 & 0 \\ 0 & \tilde{p}_3^2 + \tilde{p}_4 & -\tilde{p}_3 \\ 0 & -\tilde{p}_3 & 1 \end{pmatrix}. \quad (27)$$

It is evident that $\dot{V}_1(x)$ is negative-definite under the condition that Q_1 in (26) and Q_2 in (27) are both positive definite. Let $\rho_{\min}(\cdot)$ and $\rho_{\max}(\cdot)$ denote the least and largest eigenvalues. $\dot{V}_1(x)$ and $V_1(x)$ are further confined as follows

$$\dot{V}_1(x) \leq -\frac{1}{\sqrt{|r_1|}} \rho_{\min}(Q_1) \|x\|_2^2 - \rho_{\min}(Q_2) \|x\|_2^2, \quad (28)$$

and

$$\rho_{\min}(Q) \|x\|_2^2 \leq V_1(x) \leq \rho_{\max}(Q) \|x\|_2^2. \quad (29)$$

As a consequence, it renders

$$\begin{aligned} \dot{V}_1(x) &\leq -\frac{\sqrt{\rho_{\min}(Q)V_1(x)\rho_{\min}(Q_1)}}{\rho_{\max}(Q)} - \frac{\rho_{\min}(Q_2)}{\rho_{\max}(Q)}V_1(x) \\ &\leq -\theta\sqrt{V_1(x)}, \end{aligned} \quad (30)$$

for $\theta > 0$, and

$$\theta = \frac{\sqrt{\rho_{\min}(Q)\rho_{\min}(Q_1)}}{\rho_{\max}(Q)}. \quad (31)$$

Integrating (30) from the both sides, we are led to

$$\sqrt{V_1(x(t))} \leq \sqrt{V_1(x(0))} - 0.5\theta t. \quad (32)$$

Consequently, the sliding variable $r(t)$ turns to zero within finite time T_f , which has an explicit upper bound.

$$T_f \leq 2\theta^{-1}\sqrt{V_1(x(0))}. \quad (33)$$

This completes the proof.

The optimal controller (18) consists of four items, where the proportion parts $p_1\text{sign}(r(t))\sqrt{|r(t)|}$ and $p_3r(t)$ seek to drive $r(t)$ to zero quickly, and the integral parts $p_2\int_0^t \text{sign}(r(s))ds$ and $p_4\int_0^t r(s)ds$ aim to improve the tracking accuracy and reduce the vibrations near the equilibrium point. Note that $p_1\text{sign}(r(t))\sqrt{|r(t)|}$ and $p_2\int_0^t \text{sign}(r(s))ds$ are introduced to enhance the robustness against uncertainties and disturbances when the load current varies. Moreover, with the introduce of these items, the chattering is concealed in high frequency by choosing appropriate value of p_i . To this effect, p_1 and p_2 are selected to be relatively large, p_3 is selected to be small, and p_4 is selected to be large for fast response, good robustness, and high tracking accuracy (Levant and Levantovsky, 1993).

3.2. Optimal tracking control policy

In this part, the optimal tracking control policy is investigated. Consider the affine nonlinear system

$$\dot{x} = f_1(x) + f_2(x)u, \quad (34)$$

where $x \in \mathbb{R}^n$ is the system state, $f_1(\cdot)$ and $f_2(\cdot)$ are bounded nonlinear certain function. Let $x_d, u_d \in \mathbb{R}^n$ refer to the desired system state and desired input. The state space equation of desired system is thus described by

$$\dot{x}_d = f_1(x_d) + f_2(x_d)u_d. \quad (35)$$

Let $x_e, u_e \in \mathbb{R}^n$ refer to the state error and input error with $x_e = x - x_d, u_e = u - u_d$. The following state space expression of x_e is derived

$$\dot{x}_e = f_{1e}(x) + f_{2e}(x)u_d + f_2(x)u_e, \quad (36)$$

where

$$\begin{aligned} f_{1e}(x) &= f_1(x) - f_1(x_d), \\ f_{2e}(x) &= f_2(x) - f_2(x_d), \end{aligned} \quad (37)$$

In view of (36), it is worth noting that the optimal tracking control law on u consists of two parts, u_d and u_e . u_d is the steady-state tracking controller, which drives the system states to the equilibrium point of system (35), while u_e is the optimal controller to make sure the system states are regulated optimally (Zhang et al., 2012). Obviously,

$$u_d = (f_2(x_d))^{-1*}(\dot{x}_d - f_1(x_d)). \quad (38)$$

Herein, $(f_2(x_d))^{-1*}$ refers to the pseudo-inverse of $f_2(x_d)$.

Let u_e^* denote the optimal input of u_e . Based on optimal control theories, the infinite-horizon performance index function is set up as

$$J(u_e(v), x_e(v)) = \int_t^\infty \phi(u_e(v), x_e(v))dv, \quad (39)$$

where $\phi(u_e, x_e) = x_e^T L x_e + u_e^T K u_e$, and $L, K \in \mathbb{R}^n$ are some positive definite matrices with appropriate dimensions. As a result, the following Hamilton equation is obtained

$$H(x_e, u_e, J_{x_e}) = \phi(u_e, x_e) + J_{x_e}^T(f_{1e} + f_{2e}u_d + f_2(x)u_e), \quad (40)$$

where $J_{x_e} = \frac{\partial J}{\partial x_e}$, which denotes the partial derivative of the performance index function with respect of x_e . It is readily to get the HJB equation

$$\min \{H(x_e, u_e, J_{x_e}^*)\} = 0, \quad (41)$$

and the optimal performance index function

$$J^*(u_e(v), x_e(v)) = \min \int_t^\infty \phi(u_e(v), x_e(v))dv. \quad (42)$$

In this vein, it is useful to make the assumption that $J^*(\cdot)$ is achievable and continuously differentiable. Taking the partial derivative of (41) with respect to u_e , the optimal control law is obtained

$$u_e^* = -\frac{1}{2}K^{-1}f_2^T(x)J_{x_e}^*, \quad (43)$$

with $J_{x_e}^* = \frac{\partial J^*}{\partial x_e}$. Detailed procedures are demonstrated in Algorithm 1.

Algorithm 1 PI strategy

Require: variables: $i, i_{\max}, u_e^{(i)}, J^{(i)}, \sigma$, which denote the iteration index, maximum iteration number, i -th iteration tracking control law, i -th iterative performance index function and computational precision, respectively.

Ensure: $u_e^{(i)} \rightarrow u_e^*, J^{(i)} \rightarrow J^*$ when $i \rightarrow i_{\max}$.

Initialize the admissible control law $u_e^{(0)}$ for $i = 0$.

while $\|J^{(i+1)} - J^{(i)}\| \geq \sigma$ **do**

 Calculate $J^{(i+1)}$ by

$$\phi(u_e^{(i)}, x_e) + (J_{x_e}^{(i+1)})^T(f_{1e} + f_{2e}u_d + f_2(x)u_e^{(i)}) = 0$$

 Update the control policy by

$$u_e^{(i+1)} = -\frac{1}{2}K^{-1}f_2^T(x)J_{x_e}^{(i)}$$

 Continue the iterations

$i = i + 1$

if $i > i_{\max}$ **then**

 the optimal tracking control law is not achieved within $i_{\max}$ iterations and a larger value of $i_{\max}$ can be explored by trial and errors for further iterations.

end if

end while

Finally, the optimal tracking controller is obtained

$$u^* = (f_2(x_d))^{-1*}(\dot{x}_d - f_1(x_d)) - \frac{1}{2}K^{-1}f_2^T(x)J_{x_e}^*. \quad (44)$$

3.3. ADP implementation

As stated in Algorithm 1, the i th iteration performance index function $J^{(i)}$ is calculated by solving the complex HJB equation (40) iteratively, which requires large computation and remains challenging. In this subsection, we shall set up the ADP

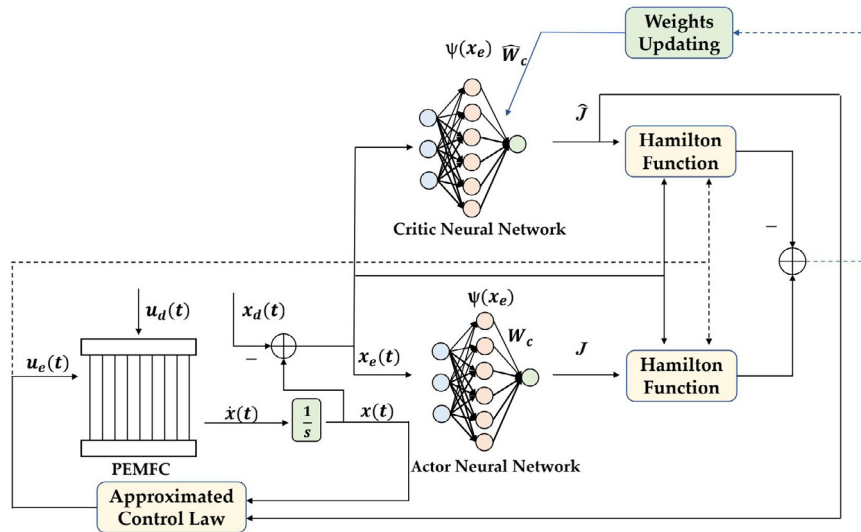


Fig. 3. Structure of ADP based optimal tracking controller.

based controller to approximate the optimal tracking control law obtained in (44). More specifically, Fig. 3 shows the structure of ADP based optimal tracking controller, where two simple three-layer NNs are employed as actor NN and critic NN, respectively. The performance index function of the actor NN is

$$J(x_e) = W_c^T \psi(x_e) + \zeta \quad (45)$$

where $W_c \in \mathbb{R}^n$ denotes the ideal weight vector of the actor NN, $\psi(\cdot)$ denotes the activation function, $\zeta \in \mathbb{R}^n$ denotes the approximate error between the actual controller and the NN controller. Based on the universal approximation property (Park and Sandberg, 1991), there must exist a NN such that ζ is sufficient small.

Define $\psi_{x_e} = \frac{\partial \psi}{\partial x_e}$, $\zeta_{x_e} = \frac{\partial \zeta}{\partial x_e}$. The partial derivative of (45) is thus expressed as

$$J_{x_e} = \psi_{x_e}^T W_c + \zeta_{x_e}. \quad (46)$$

In view of (40), the following HJB equation is derived

$$H(x_e, u_e, W_c) = \phi(x_e, u_e) + W_c^T \psi_{x_e} \dot{x}_e = e_r, \quad (47)$$

with $e_r = -\zeta_{x_e} \dot{x}_e$ being the residual error of the actor NN.

In what follows, the critic NN is introduced to approximate function (45). Let $\hat{W}_c \in \mathbb{R}^n$ refer to the real weight vector of the critic NN, and $\tilde{W}_c \in \mathbb{R}^n$ refer to the weight estimation error therein with $\tilde{W}_c = W_c - \hat{W}_c$. The approximate performance index function is described by

$$\hat{J}(x_e) = \hat{W}_c^T \psi(x_e). \quad (48)$$

Similarly, its partial derivative function with respect to x_e is computed by

$$\hat{J}_{x_e} = \psi_{x_e}^T \hat{W}_c. \quad (49)$$

The approximate Hamilton function is consequently derived as

$$H(x_e, u_e, \hat{W}_c) = \phi(x_e, u_e) + \hat{W}_c^T \psi_{x_e} \dot{x}_e = e_c. \quad (50)$$

The estimation error e_c of the HJB equation is

$$e_c = e_r - \hat{W}_c^T \psi_{x_e} \dot{x}_e. \quad (51)$$

Let E_c refer to the squared estimation error with $E_c = \frac{1}{2} e_c^T e_c$. Hence, it leads to seek for the optimal weight updating law such

that the squared estimation error E_c is minimized. By virtue of the classical gradient descent approach (Tseng and Yun, 2009), the weight updating law is generated as follows

$$\begin{aligned} \dot{\hat{W}}_c &= -\alpha \frac{\partial E_c}{\partial \hat{W}_c} = -\alpha \frac{\partial E_c}{\partial e_c} \cdot \frac{\partial e_c}{\partial \hat{W}_c} \\ &= -\alpha e_c \psi_{x_e} \dot{x}_e \\ &= -\alpha e_c \dot{\psi}(x_e), \end{aligned} \quad (52)$$

with the learning rate $\alpha > 0$, and $\dot{\psi}(x_e)$ being the variation rate of $\psi(x_e)$. Moreover, the corresponding optimal feedback control policy is derived as

$$u_e^* = -\frac{1}{2} K^{-1} f_2^T(x) (\psi_{x_e}^T W_c + \zeta_{x_e}), \quad (53)$$

which is further approximated by

$$u_e^* = -\frac{1}{2} K^{-1} f_2^T(x) \psi_{x_e}^T \hat{W}_c. \quad (54)$$

In the next, we shall prove that the approximated ADP controller converges to the real optimal tracking controller, in other words, the estimation errors of NN weights are bounded. To this end, the following assumptions are useful.

Assumption 3. The variation rate of the activation function is norm-bounded by some positive constant

$$\|\dot{\psi}(x_e)\| \leq \psi_M. \quad (55)$$

Assumption 4. The nonlinear functions $f_1(\cdot)$, $f_2(\cdot)$, $f_{1e}(\cdot)$, $f_{2e}(\cdot)$ are locally Lipschitz continuous and satisfy

$$\begin{cases} \|f_{1e}\| \leq \Gamma_{1e} \|x_e\|, \\ \|f_2\| \leq \Gamma_2, \\ \|f_{2e}\| \leq \Gamma_{2e}, \\ \|u_d\| \leq \gamma \end{cases} \quad (56)$$

for some positive constants Γ_{1e} , Γ_2 , Γ_{2e} , γ .

Assumption 3 guarantees that the update law of the real weight vector in critic NN, $\hat{W}_c$ is bounded, while Assumption 4 is required to simplify the ADP-based controller synthesis. Consequently, it yields the following theorem.

Theorem 2. Consider the nonlinear target system (34) and the ADP based approximate controller in Fig. 3 with

$$u^* = u_d - \frac{1}{2} K^{-1} f_2^T(x) \psi_{x_e}^T \hat{W}_c, \quad (57)$$

where u_d is defined in (38), and the weight updating law is

$$\dot{\hat{W}}_c = -\alpha e_c \dot{\psi}(x_e). \quad (58)$$

The tracking errors are convergent to zero if $x_e(t)$ lies outside the compact set $\|\Omega\| \leq \frac{\delta^2}{2(\rho_{\min}(L)-1-\Gamma_{1e}^2-0.5\Gamma_{2e}^2)}$, and the following condition holds

$$\begin{cases} \rho_{\min}(L) \geq 1 + \Gamma_{1e} + \frac{1}{2} \Gamma_{2e}^2, \\ \rho_{\min}(K) \geq \frac{1}{2} \Gamma_{2e}^2. \end{cases} \quad (59)$$

Proof. Construct a Lyapunov function

$$V_2(x) = \frac{1}{2} x_e^T x_e + J^*(x_e), \quad (60)$$

whose variate rate is computed by

$$\dot{V}_2(x) = x_e^T \dot{x}_e + (J_{x_e}^*(x_e))^T \dot{x}_e. \quad (61)$$

In view of (41), above equation is further equivalent to

$$\begin{aligned} \dot{V}_2 &= x_e^T (f_{1e} + f_{2e} u_d + f_2(x) u_e) \\ &\quad + (J_{x_e}^*(x_e))^T (f_{1e} + f_{2e} u_d + f_2(x) u_e) \\ &= x_e^T (f_{1e} + f_{2e} u_d + f_2(x) u_e) - \phi(u_e, x_e) \\ &= x_e^T (f_{1e} + f_{2e} u_d + f_2(x) u_e) - x_e^T L x_e - u_e^T K u_e. \end{aligned} \quad (62)$$

In terms of the Young's inequality, the upper bound of $\dot{V}_2$ is obtained

$$\begin{aligned} \dot{V}_2 &\leq \|x_e\|^2 + \Gamma_{1e} \|x_e\|^2 + \frac{1}{2} \|u_d\|^2 + \frac{1}{2} \Gamma_{2e}^2 \|x_e\|^2 + \frac{1}{2} \Gamma_{2e}^2 \|u_e\|^2 \\ &\quad - \rho_{\min}(L) \|x_e\|^2 - \rho_{\min}(K) \|u_e\|^2 \\ &\leq - \left(\rho_{\min}(L) - 1 - \Gamma_{1e} - \frac{1}{2} \Gamma_{2e}^2 \right) \|x_e\|^2 \\ &\quad - \left(\rho_{\min}(K) - \frac{1}{2} \Gamma_{2e}^2 \right) \|u_e\|^2 + \frac{1}{2} \gamma^2. \end{aligned} \quad (63)$$

As a result, $\dot{V}_2 \leq 0$ if $x_e(t)$ lies outside the compact set confined to

$$\|\Omega\| \leq \frac{\delta^2}{2(\rho_{\min}(L) - 1 - \Gamma_{1e}^2 - 0.5\Gamma_{2e}^2)}, \quad (64)$$

and the condition (59) holds. The rest of proof can be completed readily from Lyapunov stability theories and NN theories, thus is omitted herein.

4. Simulation results

In this section, the implementation of OER controller in PEMFC systems is presented. The nominal values of the parameters in PEMFC systems are obtained in Laghrouche et al. (2015) and Talj et al. (2010), as showed in Table 3. Numerical examples are demonstrated to show the advantages of our proposed OER control scheme.

Let T_{atm} denote the ambient atmospheric temperature of PEMFC. In our simulation, the value of T_{atm} is uncertain, varying from [268.34 – 357.78] (K). In the first place, we initialize the system states as

$$x_0 = [60000, 55000, 80, 20000]^T.$$

Table 3

Nominal values of system parameters.

Symbol	Value	Explanation
n	90	Number of cells in stack
R	8.314	Universal gas constant (J/(mol K))
T_{fc}	353.15	Temperature of the fuel cell (K)
F	96485	Faraday constant (C/mol)
p_{atm}	1.01325×10^5	Atmospheric pressure (Pa)
ρ_a	1.23	Air density (kg/m ³)
p_{sat}	3169	Saturation pressure at T_{atm} (Pa)
$p_{sat, T_{fc}}$	47360	Saturation pressure at T_{fc} (Pa)
V_{ca}	0.0015	Cathode volume (m ³)
V_{sm}	0.003	Supply manifold volume (m ³)
$k_{ca, in}$	0.3629×10^{-5}	Cathode inlet constant (kg/(Pa s))
$k_{ca, out}$	0.76×10^{-4}	Cathode outlet constant (kg/(Pa s))
$k_{sm, out}$	0.3629×10^{-5}	Supply manifold outlet constant (kg/(Pa s))
$x_{O_2, atm}$	0.23	Oxygen mass fraction
ω_{atm}	0.01	Humidity ratio
γ	1.4	Ratio of specific heats of air
M_a	28.9644×10^{-3}	Air molar mass (kg/mol)
M_{O_2}	32×10^{-3}	Oxygen molar mass (kg/mol)
M_{N_2}	28×10^{-3}	Nitrogen molar mass (kg/mol)
C_p	1004	Specific heat capacity of air (J/(kg K))
J_{cp}	671.9×10^{-5}	Compressor inertia (kg m ²)
η_{cp}	0.8	Compressor efficiency
η_{cm}	0.98	Motor mechanical efficiency
k_t	0.31	Motor Constant (N m/A)
k_v	0.0153	Motor Constant (V/(rad/s))
f	0.00136	Motor friction (V/(rad/s))
$V_{cp/tr}$	5×10^{-4}	Compressor volume per turn (m ³ /tr)

Through trial-and-error method, the parameters of the HOSM controller in the external control loop are set as $H = 1$, $\tilde{p}_1 = 20$, $\tilde{p}_2 = 20$, $\tilde{p}_3 = 3$, $\tilde{p}_4 = 2000$. Consequently, the tuning parameters are

$$\begin{cases} p_1 = 20, \\ p_2 = 20, \\ p_3 = 3, \\ p_4 = 2000. \end{cases} \quad (65)$$

Several offline open-loop system experiments are conducted in various experimental environments such that the optimal value of p_{ca}^* and p_{sm}^* with respect to ω_{cp}^* are achieved. Define the measurable errors $e_1 = p_{ca} - p_{ca}^*$, $e_4 = p_{sm} - p_{sm}^*$, and the tracking error $e_3 = \omega_{cp} - \omega_{cp}^*$.

Assume the initial weights of the critic NN are $\hat{W}_{c0} = [30, 5, 18, 40, 15, 60]^T$, and the activation function is $\psi(e) = [e_1^2, e_3^2, e_4^2, e_1 e_3, e_1 e_4, e_3 e_4]^T$. Based on Theorem 2, let $L = [1 \ 1 \ 1]^T$,

$$K = \begin{cases} 0.8, & |e_3| > 0.3, \\ 6, & |e_3| \leq 0.3. \end{cases} \quad (66)$$

Note that L is a constant vector, while K varies during the experiment. The learning rate α is 0.005. The optimal OER is 2.05, that is $\lambda_{O_2}^* = 2.05$. The current variate rate is assumed no larger than 10 A/s. We choose the sinusoid as the load current with

$$I_{st} = 50 \sin\left(\frac{\pi}{20} t\right) + 100. \quad (67)$$

As such, we examine the robustness and performance of the proposed OER control law compared with others.

The OER response comparison is indicated in Fig. 4, where different OER control policies, namely the fuzzy logic control-fuzzy self-tuning (FLC-FST) PID controller (Zhu et al., 2020a), the FSTPID

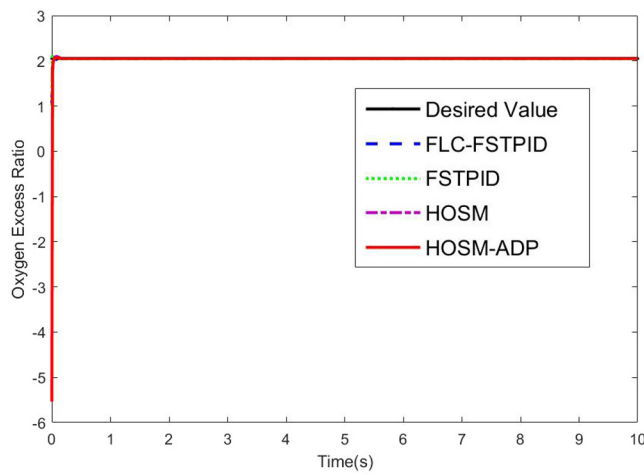
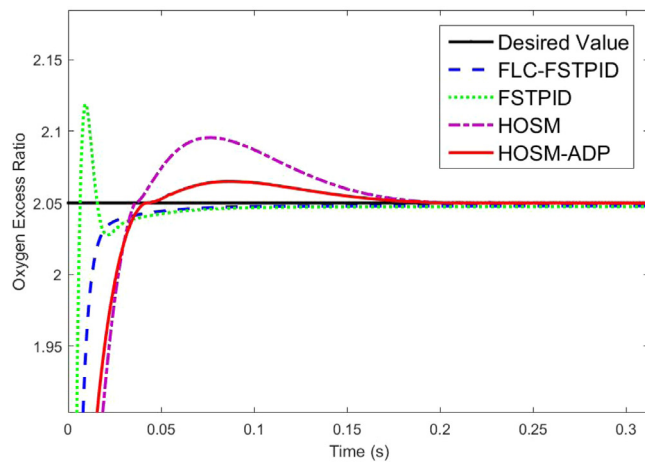
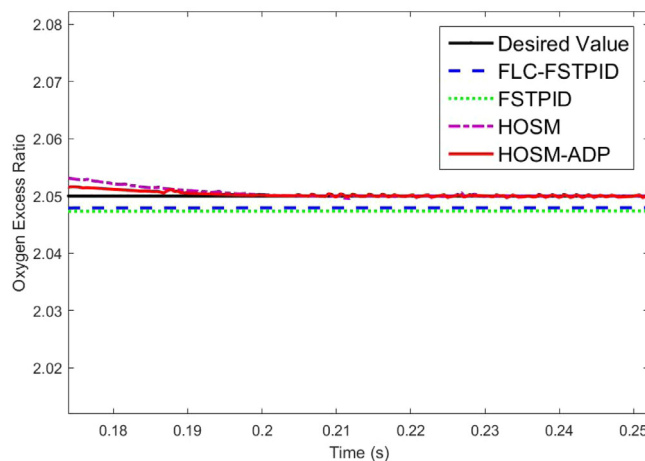


Fig. 4. OER response comparison.



(a) Enlarged view at 0.15s



(b) Enlarged view at 0.2s

Fig. 5. Enlarged view of OER response.

controller (Baroud et al., 2017), the HOSM controller (Rakhtala et al., 2015), and our proposed HOSM-ADP controller, are compared. The enlarged views are as depicted in Fig. 5, from which we conclude that though FLC-FSTPID and FSTPID controllers demonstrate faster response than the HOSM and HOSM-ADP controllers,

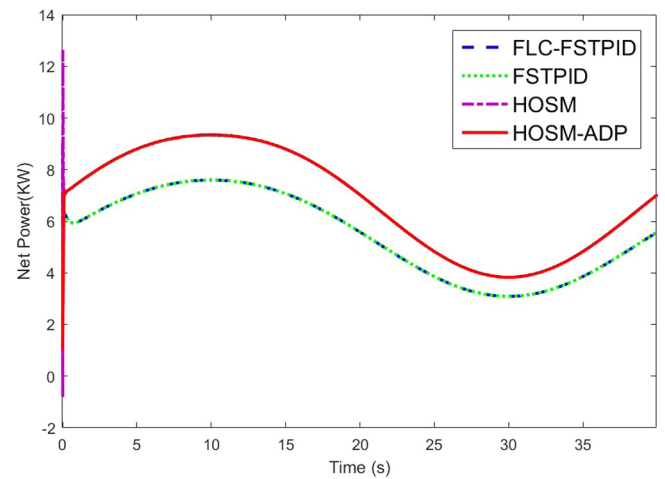


Fig. 6. Net power comparison.

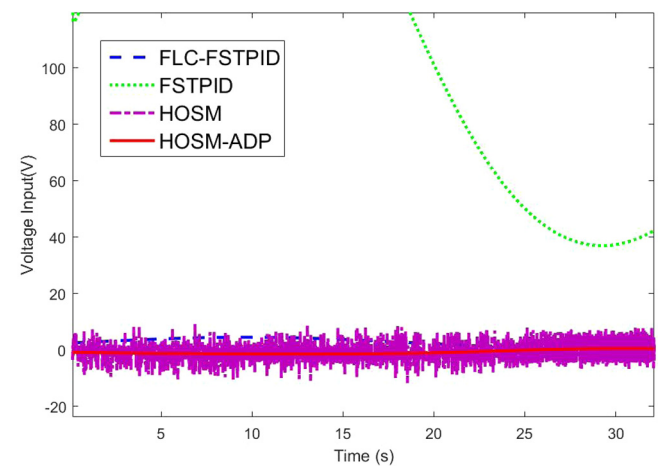


Fig. 7. Voltage response comparison.

the latter achieve higher tracking accuracy. In addition, the overshoot of the HOSM-ADP controller is much less than that of the HOSM controller.

Net power of these OER control policies are also compared. As indicated in Fig. 6, the HOSM and HOSM-ADP controllers contribute to larger net power than the others. The voltage variations are compared in Fig. 7. Since that the high switching voltage is undesired for the motor of compressor, the proposed HOSM-ADP control policy achieves the best performance among the rest without any chattering. Fig. 8 shows the convergence of the critic NN weights, from which we can see that the proposed OER control policy is convergent to the real optimal tracking control policy.

5. Conclusion

In this paper, the OER regulation problem on the PEMFC system is investigated by using the double-loop cascade control structure. Our control objective is to regulate the actual OER into the desired value with the existence of dynamics and disturbances. Simulation results show that the proposed control strategy is robust when the load current varies. Moreover, the control response is fast and smooth, and the voltage input is chattering-free. With plenty of neural network toolboxes and advanced ADP algorithms, the proposed OER control strategy can be enhanced further. In the real PEMFC system, there exist

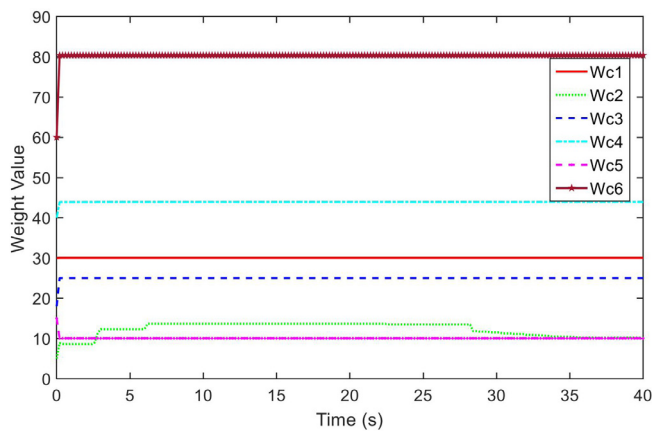


Fig. 8. The convergence of the critic NN weights.

large parametric dynamics and uncertainties without boundary information, therefore, it is of significant importance to develop refined ADP optimal control algorithms to cope with uncertainties in the future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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